Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)		
Lab exp 8	**Date: 22-08-2026**	**Enrolment No:92400133079**

CODE

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

image_path = r"D:\SEM 3\python\IMG_210.JPG"

image = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Error: Image not found!")
    exit()

histogram = cv2.calcHist(
    [image],
    [0],
    None,
    [256],
    [0, 256]
)

plt.figure(figsize=(8, 6))
plt.title('Original Image Histogram')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')
plt.plot(histogram)
plt.xlim([0, 256])
plt.grid(True)
plt.show()

equalized_image = cv2.equalizeHist(image)

plt.figure(figsize=(10, 5))

plt.subplot(1, 2, 1)
plt.title('Original Image')
plt.imshow(image, cmap='gray')
plt.axis('off')
```

```
plt.subplot(1, 2, 2)
plt.title('Equalized Image')
plt.imshow(equalized_image, cmap='gray')
plt.axis('off')

plt.tight_layout()
plt.show()

equalized_histogram = cv2.calcHist(
    [equalized_image],
    [0],
    None,
    [256],
    [0, 256]
)

plt.figure(figsize=(8, 6))
plt.title('Equalized Image Histogram')
plt.xlabel('Pixel Value')
plt.ylabel('Frequency')
plt.plot(equalized_histogram)
plt.xlim([0, 256])
plt.grid(True)
plt.show()

source_path = '/content/IMG_20230915_135931789.jpg'
reference_path = '/content/matched_image.jpg'

source_image = cv2.imread(
    source_path,
    cv2.IMREAD_GRAYSCALE
)

reference_image = cv2.imread(
    reference_path,
    cv2.IMREAD_GRAYSCALE
)

if source_image is None:
    print("Error: Source image not found!")
    exit()

if reference_image is None:
```

```
  print("Error: Reference image not found!")
    exit()

source_hist = cv2.calcHist(
    [source_image],
    [0],
    None,
    [256],
    [0, 256]
)

reference_hist = cv2.calcHist(
    [reference_image],
    [0],
    None,
    [256],
    [0, 256]
)

source_hist /= source_hist.sum()
reference_hist /= reference_hist.sum()

source_cdf = source_hist.cumsum()
reference_cdf = reference_hist.cumsum()

mapping = np.interp(
    source_cdf,
    reference_cdf,
    range(256)
)

matched_image = mapping[source_image]

matched_image = matched_image.astype(np.uint8)

plt.figure(figsize=(12, 6))

plt.subplot(1, 3, 1)
plt.title('Source Image')
plt.imshow(source_image, cmap='gray')
plt.axis('off')

plt.subplot(1, 3, 2)
```

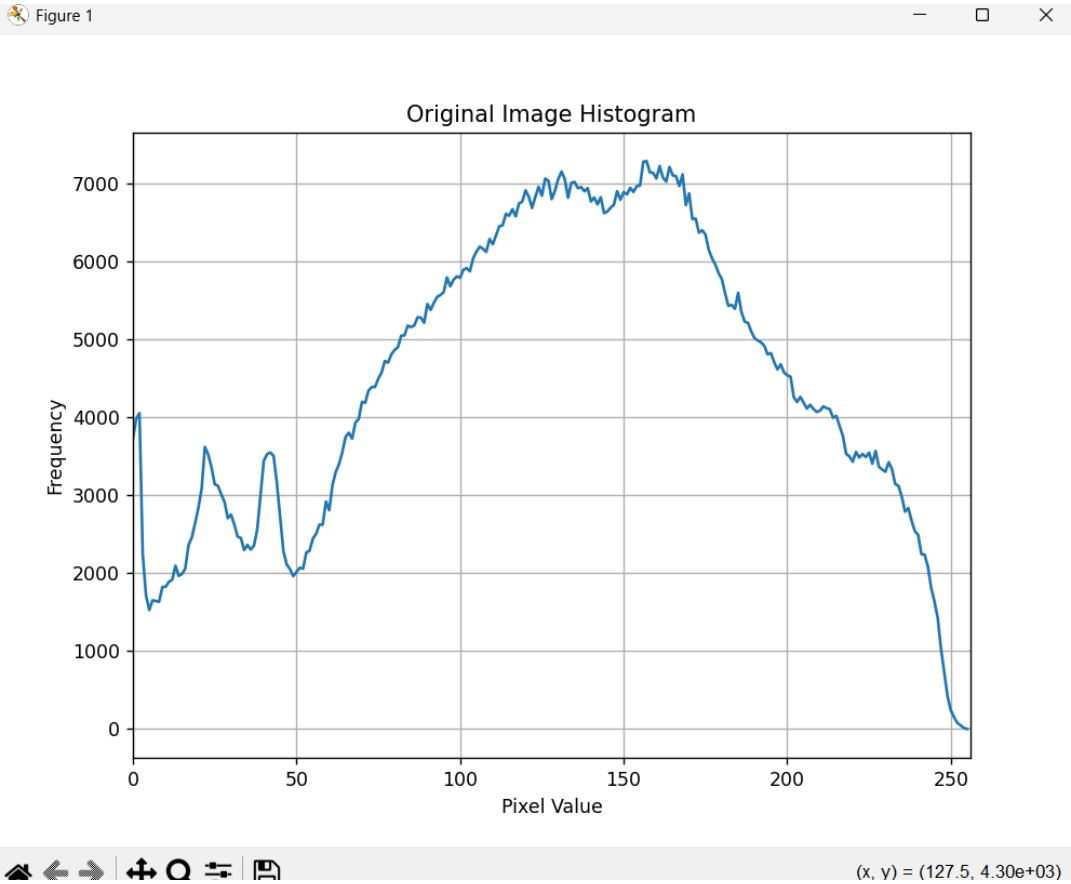
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```
plt.title('Reference Image')
plt.imshow(reference_image, cmap='gray')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.title('Matched Image')
plt.imshow(matched_image, cmap='gray')
plt.axis('off')

plt.tight_layout()
plt.show()
```

OUTPUT



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